

Pizza Sales Analysis using MySQL

PIZZA



This project analyzes pizza sales data using MySQL by solving real-world business questions. It demonstrates SQL skills such as joins, aggregate functions, subqueries, and window functions to generate meaningful business insights.



SQL Concepts Used:

SELECT
WHERE
GROUP BY
ORDER BY
LIMIT
INNER JOIN
Aggregate Functions
Subqueries
Window Functions
RANK()
OVER()

Objective

The objective of this project is to analyze pizza sales data using SQL and answer business-related questions that help understand customer purchasing behavior, sales performance, and revenue trends.

The project uses four related tables:

Table Name	Description
Orders	Stores order date and order time
Order_Details	Stores quantity of pizzas ordered
Pizzas	Stores pizza size and price
Pizza_Types	Stores pizza names and categories

Tools Used:

- MySQL Workbench
- SQL
- Canva (Presentation Design)

Retrieve the total number of orders placed.

```
SELECT
    COUNT(order_id) AS Total_Orders
FROM
    orders;
```

Output:

Result Grid	
	Total_Orders
▶	21350



Calculate the total revenue generated from pizza sales

```
SELECT
    ROUND(SUM(quantity * price), 2) AS Total_Revenue
FROM
    order_details od
    JOIN
    pizzas AS p ON p.PIZZA_ID = od.PIZZA_ID
;
```

Output:

	Total_Revenue
▶	817860.05

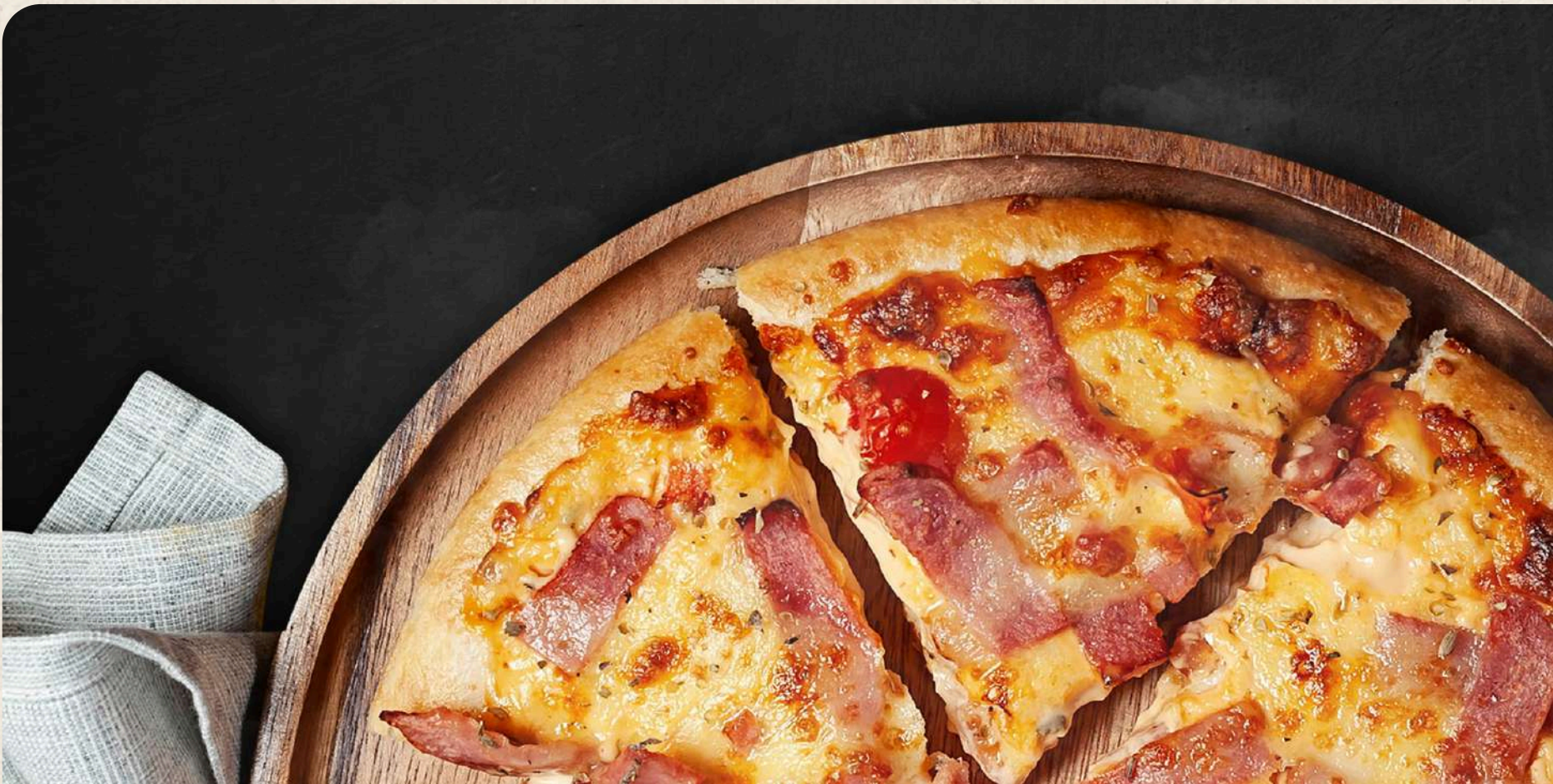


Identify the highest-priced pizza.

```
SELECT
    MAX(price) AS Highest_priced_pizza
FROM
    PIZZAS;
```

Output:

	Highest_priced_pizza
▶	35.95





Identify the most common pizza size ordered

```
SELECT
    size, COUNT(order_details_id) AS order_count
FROM
    pizzas AS p
    JOIN
    order_details AS o ON p.pizza_id = o.pizza_id
GROUP BY size
ORDER BY order_count DESC
LIMIT 1;
```

Output:

	size	order_count
►	L	18526

List the top 5 most ordered pizza types along with their quantities.

```
SELECT
    pizza_type_id AS Pizzas_Types, SUM(quantity) AS Orders
FROM
    pizzas AS p
    JOIN
    order_details AS o ON p.pizza_id = o.pizza_id
GROUP BY Pizzas_types
ORDER BY Orders DESC
LIMIT 5;
```

Output:

	Pizzas_Types	Orders
▶	classic_dlx	2453
	bbq_ckn	2432
	hawaiian	2422
	pepperoni	2418
	thai_ckn	2371



Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT
  category, SUM(quantity) AS quantity
FROM
  pizza_types AS pt
  JOIN
  pizzas AS p ON pt.pizza_type_id = p.pizza_type_id
  JOIN
  order_details AS o ON o.pizza_id = p.pizza_id
GROUP BY category
ORDER BY quantity DESC;
```

Output:

	category	quantity
►	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050



Determine the distribution of orders by hour of the day.

```
SELECT
    HOUR(order_time) AS Hours, COUNT(order_id) AS order_counts
FROM
    orders
GROUP BY Hours;
```

Output:

	Hours	order_counts
▶	11	1231
	12	2520
	13	2455
	14	1472
	15	1468
	16	1920
	17	2336
	18	2399
	19	2009
	20	1642
	21	1198
	22	663
	23	28
	10	8
	9	1



Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

Output:

	category	COUNT(name)
▶	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9

Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    ROUND(AVG(quantity), 0) AS avg_pizzas_order
FROM
    (SELECT
        order_date, SUM(quantity) AS quantity
    FROM
        orders AS o
    JOIN order_details AS od ON o.Order_id = od.Order_id
    GROUP BY order_date) AS order_quantity;
```

Output:

	avg_pizzas_order
▶	138

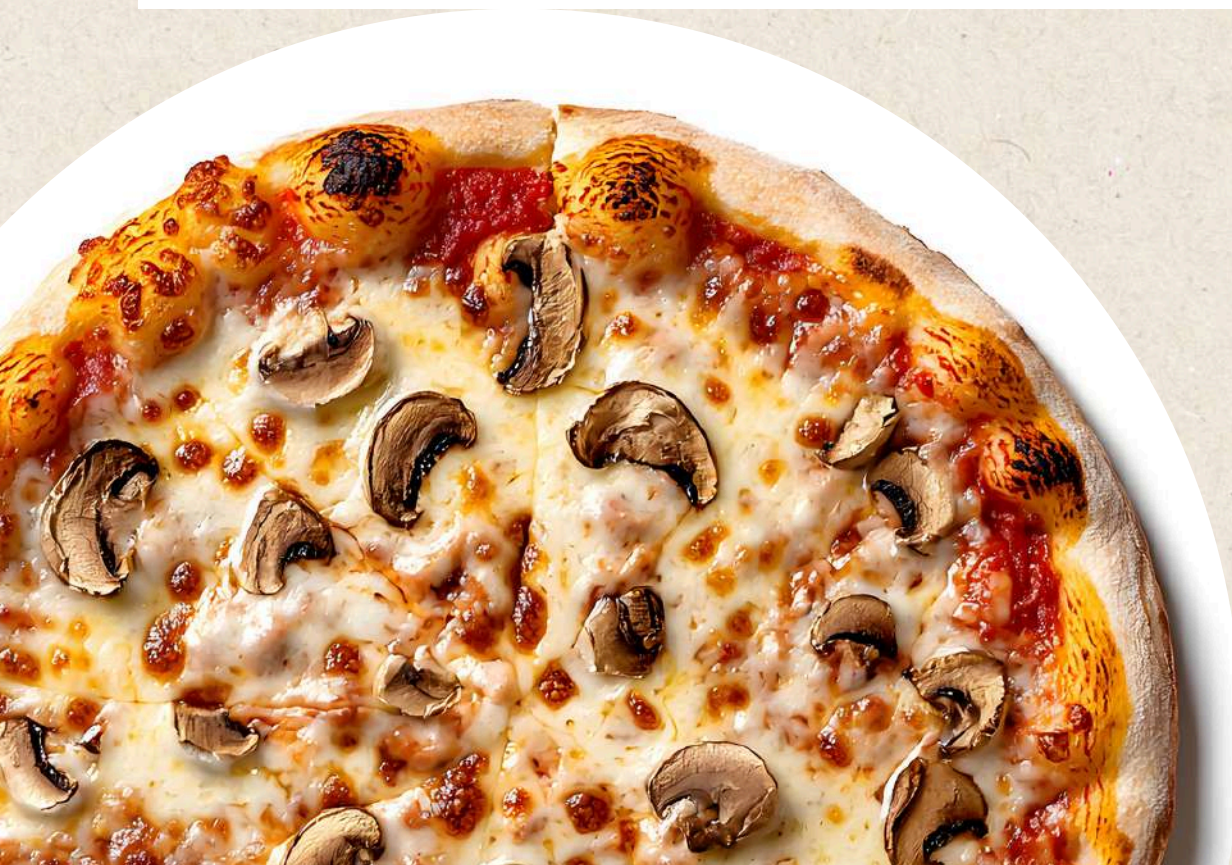


Determine the top 3 most ordered pizza types based on revenue.

```
SELECT
    pizza_type_id AS pizza_types,
    SUM(quantity * price) AS Total_Revenue
FROM
    order_details AS od
    JOIN
    pizzas AS p ON od.pizza_id = p.pizza_id
GROUP BY pizza_types
ORDER BY Total_Revenue DESC
LIMIT 3;
```

Output:

	pizza_types	Total_Revenue
▶	thai_chn	43434.25
	bbq_chn	42768
	cali_chn	41409.5



Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT
    SUM(od.quantity * p.price)
FROM
    order_details od
    JOIN
    pizzas p ON od.pizza_id = p.pizza_id;

SELECT
    pt.name AS pizza_type,
    ROUND(SUM(od.quantity * p.price) * 100 / (SELECT
        SUM(od2.quantity * p2.price)
        FROM
            order_details od2
            JOIN
            pizzas p2 ON od2.pizza_id = p2.pizza_id),
        2) AS revenue_percentage
FROM
    order_details od
    JOIN
    pizzas p ON od.pizza_id = p.pizza_id
    JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY revenue_percentage DESC;
```

Output:

	pizza_type	revenue_percentage
►	The Thai Chicken Pizza	5.31
	The Barbecue Chicken Pizza	5.23
	The California Chicken Pizza	5.06
	The Classic Deluxe Pizza	4.67
	The Spicy Italian Pizza	4.26
	The Southwest Chicken Pizza	4.24
	The Italian Supreme Pizza	4.09
	The Hawaiian Pizza	3.95
	The Four Cheese Pizza	3.95
	The Sicilian Pizza	3.78
	The Pepperoni Pizza	3.69
	The Greek Pizza	3.48
	The Mexicana Pizza	3.27
	The Five Cheese Pizza	3.19
	The Pepper Salami Pizza	3.12
	The Italian Capocollo Pizza	3.07
	The Vegetables + Vegetable...	2.98
	The Prosciutto and Arugula ...	2.96
	The Napolitana Pizza	2.95



Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
select name, revenue from
(select category, name, revenue ,
rank() over(partition by category order by revenue desc) as rn
from
(select category, name, sum(quantity * price) as revenue
from pizza_types as pt join pizzas as p
on pt.pizza_type_id = p.pizza_type_id
join order_details as od
on od.pizza_id = p.pizza_id
group by category, name) as a) as b
where rn<=3
;
```



Output:

	name	revenue
►	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5
	The Classic Deluxe Pizza	38180.5
	The Hawaiian Pizza	32273.25
	The Pepperoni Pizza	30161.75
	The Spicy Italian Pizza	34831.25
	The Italian Supreme Pizza	33476.75
	The Sicilian Pizza	30940.5

Key Insights:

- Total Orders: 21,350
- Total Revenue: \$817,860.05
- Most Ordered Pizza Size: Large
- Highest Priced Pizza: \$35.95
- Top Revenue Pizza: Thai Chicken Pizza
- Peak Order Time: 12 PM–1 PM

